Research and system implementation of smart launch based on consumer big data

Xuexia Liang
Internet Research Center
China Tobacco Guangxi Industrial Co., LTD.
Nanning, Guangxi, China
07068@gxzy.cn

Chao Deng
Internet Research Center
China Tobacco Guangxi Industrial Co., LTD.
Nanning, Guangxi, China
dengch021@hotmail.com

Bin Lu*
Internet Research Center
China Tobacco Guangxi Industrial Co., LTD.
Nanning, Guangxi, China
931926529@qq.com

Hui Huang
Cigarette Marketing Center
Guangxi Zhuang Autonomous Region Tobacco Corporation Guigang City Company
Guigang, Guangxi, China
80850058@qq.com

Jianping Zhu
Cigarette Marketing Center
Guangxi Zhuang Autonomous Region Tobacco Corporation Guigang City Company
Guigang, Guangxi, China
jpzhu@qq.com

Paiyu Li
Internet Research Center
China Tobacco Guangxi Industrial Co., LTD.
Nanning, Guangxi, China
07130@gxzy.cn

Abstract

In order to solve the problem of accurate launch of cigarette products, an cigarette smart launch system was established based on consumer big data, aiming to improve the efficiency and effect of cigarette market launch through more scientific and accurate methods. By building a big data labeling system for end consumers and using big data batch processing pipelines, dynamic assessment of the sales potential of different terminals can be achieved. Furthermore, a variety of launch modes were designed and implemented, including "by grade" launch, "grade × label" launch, "grade + label expansion" launch, and new product selection point launch strategies. Through the intuitive visual interface and convenient interactive functions, the system greatly improves the efficiency of cigarette launch strategy formulation. The system was applied in the Yulin cigarette market in Guangxi, and the delivery accuracy rate increased by 14.8%. In the application of the Jinan market, customer satisfaction increased by an average of 12%.

CCS Concepts

• **Information systems** → Information systems applications; Decision support systems; Data analytics.

Keywords

big data, cigarette market launch, consumer behavior analysis, sales potential, smart launch system

*Corresponding author.

ICCSMT 2024, October 18–20, 2024, Xiamen, China

ACM ISBN 979-8-4007-0999-9/24/10
https://doi.org/10.1145/3708036.3708073

ACM Reference Format:
Xuexia Liang, Chao Deng, Bin Lu, Hui Huang, Jianping Zhu, and Paiyu Li. 2024. Research and system implementation of smart launch based on consumer big data. In *2024 5th International Conference on Computer Science and Management Technology (ICCSMT 2024), October 18–20, 2024, Xiamen, China.* ACM, New York, NY, USA, 8 pages. https://doi.org/10.1145/3708036.3708073

1 INTRODUCTION

The market launch of cigarette products refers to the process in which tobacco commercial companies distribute products to retail terminals. It is also a crucial foundational task for these companies, as launch strategies and outcomes directly impact their economic performance. In the process of cigarette placement, it is essential to balance market consumption demand with the principle of market fairness. On the one hand, launch strategies must accurately match actual consumer needs, increasing the supply in regions with active consumption while reducing it in areas with weaker consumption capacity. On the other hand, launch efforts must consider the objective attributes of cigarette retail terminals, such as "grade," "geographic location," "business type," and "sales departments," to ensure fairness in distribution, thereby safeguarding the interests of every retailer. Given the vast variety of cigarette products available in the market, there is an urgent need to conduct research on launch strategies to achieve scientific, fair, and precise distribution. This would enable better adaptation to and fulfillment of diverse market demands, improving market efficiency and customer satisfaction.

In terms of using data analysis methods to measure and analyze cigarette launch [1], Xu Hong et al. [2] applied data mining technology and fuzzy comprehensive evaluation method to quantitatively score the sales ability of retail terminals, and optimized the allocation of supply based on the scoring results. Deng Chao et al. [3] proposed an intelligent cigarette launch model that integrates long short-term memory network and back propagation neural network, aiming to improve the accuracy of launch. Mo Yuhua et al. [4] combined the synthetic control method with the machine learning model, and used the stable sales trend of mature cigarette products to simulate and calibrate the sales trend of new cigarette products, providing a reference for the launch of new products . Liang Xuexia et al. [5] proposed an intelligent evaluation model of cigarette market operation status based on machine learning algorithm, which provides data support for launch decision-making by identifying the cigarette market operation status. Xu Ruiqi et al. [6] introduced "business district label " based on customer position, combined with demand forecasting, and constructed a precise launch model of " grade + business district " . Wei Taicheng et al. [7] realized the intelligence, digitization and precision of cigarette launch by studying the demand forecast model of each cigarette specification in the next cycle and formulating intelligent launch strategies .

Most of the above literature is based on the analysis and calculation of the company's own cigarette data, ignoring the impact of external consumer big data on cigarette launch. In addition, traditional cigarette launch strategies are often manually formulated by sales staff, which is arduous and time-consuming. The main challenge in applying big data technology to the traditional cigarette distribution process lies in the complexity of data integration. The tobacco industry's sales data, inventory data and external market data sources are complex. Integrating this data requires cross-departmental collaboration, and data cleaning and processing takes a long time. In addition, market conditions change rapidly, and traditional data update cycles are long, making it difficult to capture market dynamics in real time.Therefore, this paper starts from the multi-dimensional cigarette "people, goods, and places" and builds an cigarette smart launch system based on cigarette market consumption big data, in order to quickly evaluate the sales potential of different terminals, accurately lock in consumer demand, and improve the effect of cigarette market launch and business process efficiency in a more scientific and accurate way.

Compared with the traditional cigarette launch model, the advantages of this system are mainly reflected in the following aspects:

(1) Comprehensively use multi-dimensional labels of people, goods, and places to comprehensively evaluate the sales potential of different terminals and achieve data-driven precision launch.

(2) Use Piflow big data technology to build a data processing pipeline, automatically update the underlying data on a weekly basis, capture market trends and consumer behavior in a timely manner, provide a dynamic, real-time, and accurate data foundation for upper-level applications, and realize dynamic updates of launch strategies.

(3) Provide a variety of flexible launch modes, including "by grade", "grade × label", " grade + label extension ", new product selection and launch, etc., to meet the needs of different markets and products.

(4) Through the visual interface and convenient interactive functions, business personnel can efficiently formulate strategies and intuitively understand the effects of launch, thereby improving the efficiency of cigarette cultivation.

2 OPERATING PRINCIPLES OF INTELLIGENT CIGARETTE LAUNCH

2.1 Construction of terminal consumer big data label system

Since cigarette retail mainly focuses on offline channels, most consumers prefer to buy nearby. Taking this into account, based on the concept of "15-minute convenience circle", the analysis area is divided into several 750m × 750m business district grids [8, 9]. In addition, Internet companies can provide external consumer group portraits of designated grid longitudes and latitudes. Therefore, based on the business district, cigarette "people, goods, and places" big data label fusion can be achieved. The specific approach is shown in Figure 1. First, the analysis area is evenly divided into several 750m × 750m business district grids. All terminals are accurately mapped to the business district grids through longitude and latitude to establish a one-to-one correspondence between terminals and business districts. The crowd portrait labels of each business district are purchased from external Internet companies, including the population size, gender ratio, age structure, educational background, consumption level, average hotel accommodation price, per capita catering consumption, mobile phone price, etc. in the business district, which constitute the crowd characteristics of the business district. Web crawler technology is used to crawl Point of Interest (POI) labels from public electronic maps, such as the number of catering services, financial insurance services, hotels, real estate communities, hospitals, office buildings, shopping centers, tourist attractions, etc., to form the characteristics of the business district. At the same time, each terminal is inside the business district and will inherit the basic information of the business district, the internal crowd characteristic labels and the venue characteristic labels. Combined with the terminal's own cigarette sales records, the number of orders, code scans and the amount of each cigarette specification of each terminal are counted in a specific time period to form an cigarette sales label, thereby building a personalized and refined cigarette consumer big data label system for each terminal, so as to deeply understand the differences in cigarette consumption needs, behaviors, places and populations between different terminals.

This method mainly gathers consumer information through business district grids, and replaces massive individual consumer portraits with thousands of business district consumer portraits, while protecting the privacy and security of individual consumers, further reducing data collection and analysis costs.

2.2 Launch strategy calculation

After establishing a consumer big data label with the terminal ID as the unique identifier, the correlation between the cigarette label to be launched (the historical sales data of the cigarette specification at each retail terminal) and other labels (the "people, goods, and places" labels of each retail terminal) is calculated to analyze how other labels affect the cigarette to be launched, thereby evaluating the sales potential of different terminals. The correlation coefficient is a statistical indicator used to measure the correlation between the target variable and other variables. It reflects the strength and direction of the linear relationship between the variables. The

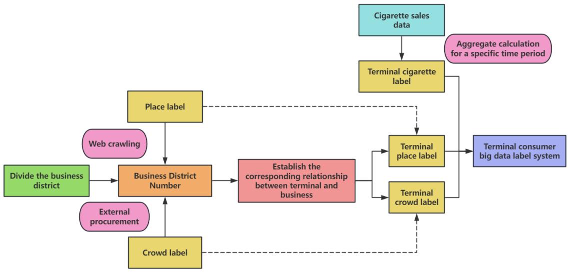

Figure 1: Terminal Consumer Big Data label Establishment Process Diagram.

value range is usually between -1 and 1, where 1 indicates a perfect positive correlation, -1 indicates a perfect negative correlation, and 0 indicates no linear relationship [10]. Positive correlation indicates that the target variable increases as the other variable increases, while negative correlation indicates that the increase of the target variable is accompanied by a decrease of the other variable. The closer the absolute value of the correlation coefficient is to 1, the stronger the relationship; the closer it is to 0, the weaker the relationship.Specifically, the calculation of the launch strategy involves the following steps:

The first step is to calculate the Pearson correlation coefficient between the tobacco to be placed and all other labels:

$$r_{A,m} = \frac{\sum_{i=1}^{N}\left(X_i - \bar{X}\right)\left(Y_{i,m} - \overline{Y_m}\right)}{\left(\sqrt{\sum_{i=1}^{N}\left(X_i - \bar{X}\right)^2}\right)\left(\sqrt{\sum_{i=1}^{N}\left(Y_{i,m} - \overline{Y_m}\right)^2}\right)} \tag{1}$$

Where N is the number of retail terminals in the analysis area, X_i is the total number of orders for cigarette A at the i-th terminal in a specific time period (the last three months in this article), $\overline{X}$ is the average order value of cigarette A at all terminals in the analysis area in a specific time period, $Y_{i,m}$ is the specific value of the m-th label at the i-th terminal, $\bar{Y}_m$ is the average value of the m-th label of all terminals in the preset area, and $r_{A,m}$is the Pearson correlation coefficient between cigarette A and the m-th label.

The second step is to normalize all labels. For positive indicators, then:

$$x_{i,m}^{*} = \frac{x_{i,m} - x_{min,m}}{x_{max,m} - x_{min,m}} \tag{2}$$

For negative indicators, then:

$$x_{i,m}^{*} = \frac{x_{max,m} - x_{i,m}}{x_{max,m} - x_{min,m}} \tag{3}$$

$x_{i,m}$is the original value of the m-th data label of the i-th terminal, $x_{max,m}$ and $x_{min,m}$ are the maximum and minimum values of the m-th data label respectively, and $x_{i,m}^{*}$is the normalized value of the m-th data label of the i-th retailer.

The third step is to perform a weighted summation of the normalized label value and the Pearson correlation coefficient corresponding to cigarette A for all terminals to obtain the ordering ability score $g_i^{'}$ of terminal i for cigarette A.

$$g_i^{'} = \sum_{m=1}^{M} r_{A,m} \times x_{i,m}^{*} \tag{4}$$

M is the total number of labels contained in the constructed consumer big data label system.

The fourth step is to consider the ordering capacity of tobacco based on the consumption level of retail terminals, and comprehensively consider the real-time inventory status and sales performance, and try to reduce the deployment of terminals with large real-time inventory and slow sales to ensure the stability of the market status. After consulting business experts and conducting a comprehensive analysis of market feedback indicators, it was finally found that the order-full rate, sales-to-order ratio and inventory-to-sales ratio have a high correlation with the status. Therefore, the three key indicators of order-full rate, sales-to-order ratio and inventory-to-sales ratio are used to comprehensively evaluate the sales and inventory status of tobacco at different terminals. The calculation formula of the indicators is as follows:

Order fulfillment rate = (order quantity/launch quantity) × 100%

Sales to order ratio = (sales volume/order volume) × 100%

Inventory-to-sales ratio = (inventory/sales) × 100%

In order to ensure the objectivity and fairness of the evaluation results, the entropy method is used to determine the weights of the above four indicators. First, the four indicators of ordering capacity score $g_i^{'}$, full order rate, sales-to-order ratio and inventory-to-sales ratio are standardized using the second step method to eliminate the dimension effect. Secondly, the weight of each indicator is determined by calculating its information entropy. The smaller the information entropy, the greater the variability of the indicator, and the higher the weight in the comprehensive evaluation [11]. The calculation formula of the information entropy and weight of each indicator is as follows:

$$e_k = -\frac{1}{ln(N)} \times \sum_{i=1}^{N}\left(\frac{x_{i,k}^{*}}{\sum_{i=1}^{N} x_{i,k}^{*}} \times ln\frac{x_{i,k}^{*}}{\sum_{i=1}^{N} x_{i,k}^{*}}\right) \tag{5}$$

$$w_k = \frac{1 - e_k}{\sum_{j=1}^{n}\left(1 - e_j\right)} \tag{6}$$

$e_k$is the entropy value of the k-th indicator, $w_k$is the weight of the k-th indicator, and n is the total number of indicators .

Finally, the weights of each indicator are combined to perform weighted summation on the standardized data to obtain $i$the final sales potential score of each terminal $g_i$for cigarette A :

$$g_i = \sum_{k=1}^{n} w_k \times x_{i,k}^{*} \tag{7}$$

$g_i$is the terminal i's final sales potential score for tobacco, $x_{i,k}^{*}$and is the value after standardized processing of any one of the four indicators of the terminal i's ordering ability score for tobacco $g_i^{'}$, full order rate, sales-to-order ratio and inventory-to-sales ratio .

The fifth step is to determine a certain launch method, calculate the total terminal score and average terminal score of different launch units in the launch method, sort the different launch units of the launch method in descending order from high to low according to the average terminal score, and calculate the total launch volume of different launch units and the average launch volume per unit based on the total launch volume S of tobacco in the next cycle:

Total unit launch volume = (total unit terminal score/total score of all terminals) × total launch volume S

Average unit household launch volume = total unit launch volume / number of unit terminals

The sixth step is that the business personnel, based on the actual market demand, round up or down the average household launch amount from high to low according to the average unit terminal score, until the total launch amount S is decomposed, then stop and the remaining units will no longer launch .

The method in this paper comprehensively examines the correlation between tobacco and many ”people, goods, and places” labels. In this way, a more refined and accurate terminal sales potential score is achieved. From the perspective of big data, when only a few labels are considered, the results may be biased. However, when the number of labels is large enough, the bias that may be caused by individual labels will be effectively diluted, thereby improving the accuracy of the evaluation results. By analyzing the correlation between a large number of labels and tobacco, the cigarette consumption behavior of each terminal can be analyzed more comprehensively to ensure that the cigarette launch strategy is more in line with the actual needs of the market.

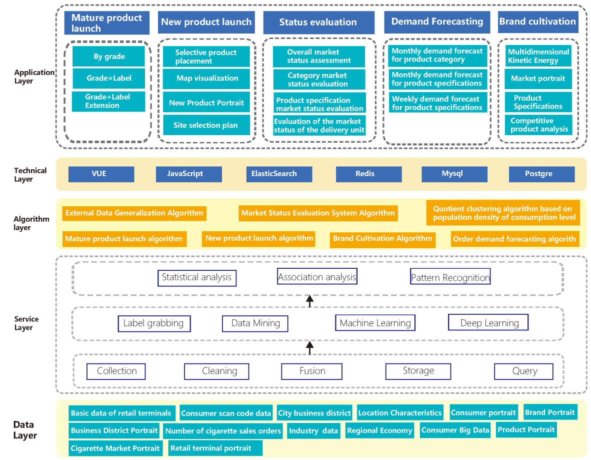

Figure 2: System Architecture Diagram.

3 SYSTEM IMPLEMENTATION

Different products have different product positioning, market segments, and consumer segments. Products need to be placed in places that meet market demand in order to improve product market performance and enhance product competitiveness. To this end, based on the rich cigarette consumer big data label system, combined with specific work business, we have designed and developed cigarette ”by grade”, ”grade × label”, ”grade + label extension”, new product selection and other launch modes using big data, artificial intelligence and other technologies, and built an cigarette smart launch system to provide digital tool support for the cigarette marketing field.

3.1 System technical architecture

The system architecture is divided into five layers, from bottom to top, namely the data layer, service layer, algorithm layer, technical layer and application layer. As shown in Figure 2, the data layer is based on the three major data label systems of ”people, goods and places” for the cigarette brand, including BB’s internal data (basic data of retail terminals, consumer scanning data, cigarette sales order data, industry downstream data) and external data (urban business districts, venue characteristics, regional economy, consumer big data, etc.). Since the cigarette launch business needs to be carried out on a weekly basis, the service layer is based on the Piflow big data pipeline distributed processing architecture, and automatically collects, cleans, calculates, aggregates and structures the underlying cigarette sales orders, scanning and inventory data on a weekly basis to ensure the real-time and accuracy of the launch results. The algorithm layer uses data mining, digital labels, label capture and machine learning technologies and methods to perform statistics, classification, clustering, association analysis and pattern recognition on consumers, cigarette brands, product specifications and consumption locations, deeply understand the cigarette consumption behavior characteristics of different consumer groups and different consumption places, and build cigarette product and consumption place, consumer group segmentation model and feature association model. The technical layer provides the system’s infrastructure support, including the front-end VUE framework, the back-end JavaScript language, and the mid-end ElasticSearch search engine [12], Redis cache database, and Mysql and Postgre relational databases, ensuring the stability and efficiency of the system. The application layer divides the system into five major functional modules, including ”by grade”, ”grade × label”, ”grade + label extension” and new product selection and release, to ensure that the system can be flexibly applied to different business needs. In addition, the system also integrates modules such as market status evaluation, cigarette market demand forecasting, and brand cultivation. By building a complete set of multi-source heterogeneous data aggregation-fusion-application processes, it realizes comprehensive analysis and decision support for the cigarette market, providing a scientific basis for the release of cigarette products.

3.2 ”By grade ” launch

”by grade” is a basic way of cigarette launch in the tobacco industry, and it is also the mainstream way. By dividing the terminals into different grades and taking the grade as the smallest launch unit, the total amount of the same specification of the terminals with the same grade is the same. When the city, product specifications and quantity are selected, the system uses data-driven intelligence to allocate the total amount of launch to each grade. Taking Nanning, Guangxi as an example, assuming that 30,000 Zhenlong(Meirenxiangcao) are placed in terminals of grades 15 to 30, the system page and launch results are shown in Figure 3.

From the results of the launch, we can clearly observe a trend that as the grade level gradually rises, the launch volume of a single household will also increase accordingly. In other words, the higher the grade level, the stronger its sales potential, and the larger the launch volume of a single household, which is consistent with objective facts.

Figure 3: "By grade" Release Interface and Results.

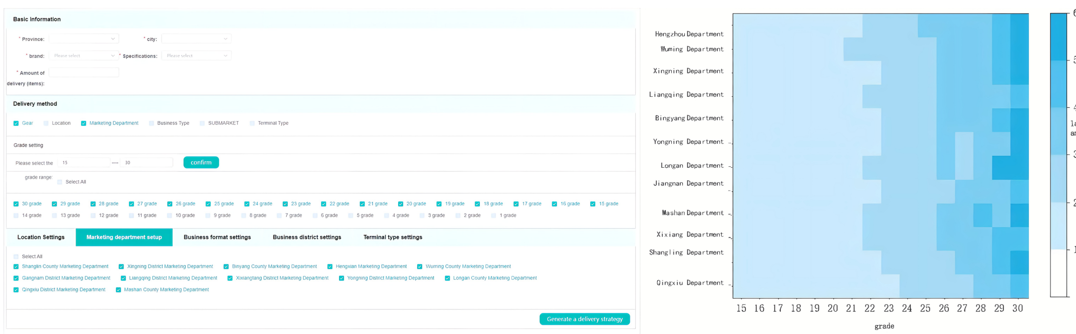

Figure 4: "Grade × label" Release Interface and Results.

3.3 " grade × label " launch

Although the "by grade" launch model guarantees the fairness of supply launch to the greatest extent, it does not take into account the differentiated supply needs of terminals in the same grade, which can easily cause an imbalance between supply and demand. Therefore, based on the "by grade" launch model, the "grade × label" launch model is proposed, where the label refers to other attribute labels assigned by the tobacco company to the terminal in addition to the basic attribute of "by grade", including geographical location, marketing department, business format, business district, terminal type and other first-level labels and subordinate second-level segmentation labels. The "grade × label" launch model takes the terminal with the same grade and label as the minimum launch unit, and the total amount of supply of the same specification for terminals with the same grade and second-level label is the same. By further refining the minimum launch unit, cigarette launch is more accurate.

Similarly, Zhenlong (Meirenxiangcao) in Nanning, Guangxi Province is still selected, and the launch volume is preset to 30,000. At the same time, the 15-30 grade and the marketing department label are selected to generate the launch strategy . Figure 4 shows the system page and launch results.

As shown in Figure 4, the distribution of the distribution volume allocated to each distribution unit in the distribution results is between 1 and 6, and the distribution volume of each marketing department is different. It basically meets the requirement that the distribution volume will increase with the increase of the grade, which is consistent with business common sense. In addition, compared with the "by grade" distribution method, the "grade × label" distribution method can not only locate the target audience more accurately, but also has a more flexible distribution strategy, which can meet the needs of different business personnel and further improve the effect of product distribution.

3.4 "grade + label extension " launch

"grade + label extension " launch mode is another evolution of grade launch. The specific idea is to split the total supply into two parts, one of which is used for terminal "by grade" to ensure fairness in terminal ordering; the other part is used for label launch, that is, terminals with corresponding labels can get additional launch, ensuring the difference in terminal ordering to meet the ordering needs of different terminals.

Taking the Zhenlong (Meirenxiangcao) launch in Nanning, Guangxi as an example, the launch volume is preset to 30,000, and the 15-30 grade and the marketing department label are selected to generate the launch strategy. In the launch ratio setting, the grade ratio is set to 70% and the marketing department ratio is set to 30%. The system page and launch results are shown in Figure 5.

Compared with the launch results of "grade × label", the launch volume of each unit is distributed between 0 and 5. This is because when delivering "grade × label", the unit score is given priority, without strict order and proportion requirements. However, in the case of "grade + label extension", the two results are merged in strict accordance with the proportion requirements to obtain the final launch result. As can be seen from Figure 5, the result still

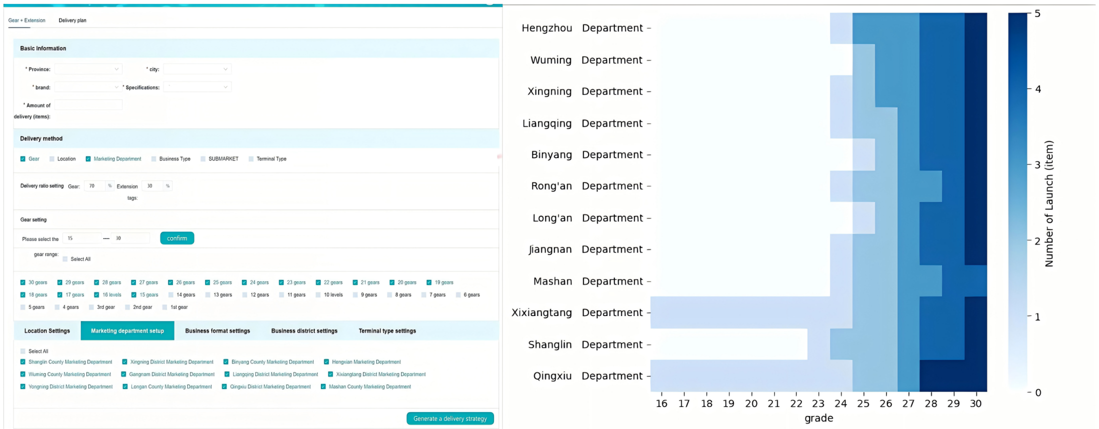

Figure 5: Grade + label Extension Release Interface and Results.

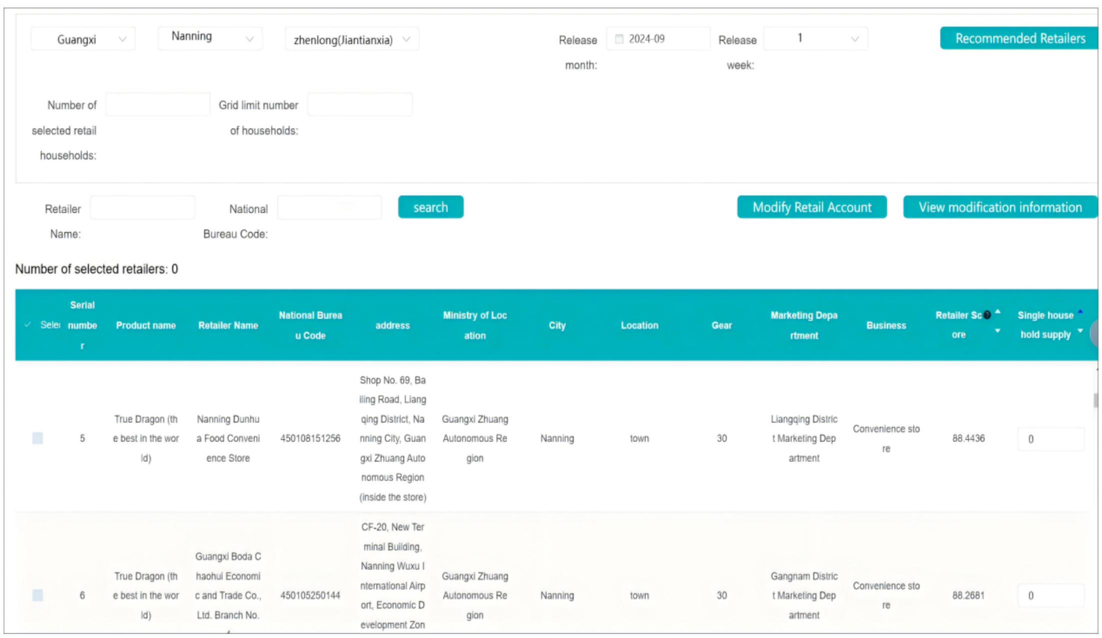

Figure 6: New Product Retail Terminal Selection Release Interface.

basically meets the trend that the launch volume will increase with the increase of grade. It is worth noting that the launch volume of the marketing department of Xixiangtang District and the marketing department of Qingxiu District is more than that of other marketing departments at lower grades. This is because when only the extended label of the marketing department is considered, the results show that these two areas are the only areas with additional expansion volume, so they show obvious differences when comprehensively considered. The advantage of this expansion model is that it allows the weight ratio to be determined by oneself, because the focus to be considered in different regions may be different.

3.5 New product launch location

Unlike mature products in the market, new products lack historical data to support their initial launch.Therefore, market competitor cigarette can be used as a reference for launch, and the TOP-N market competitor cigarette with the highest sales potential score can be selected $g_i$as the target retail terminals for new product launch,as shown in Figure 6. At the same time, after a small-scale trial launch of new products in the market, the retail terminal potential score can be dynamically updated based on market feedback data.

As shown in Figure 7, after selecting the target retail terminal, the system also provides intuitive target terminal portrait analysis,

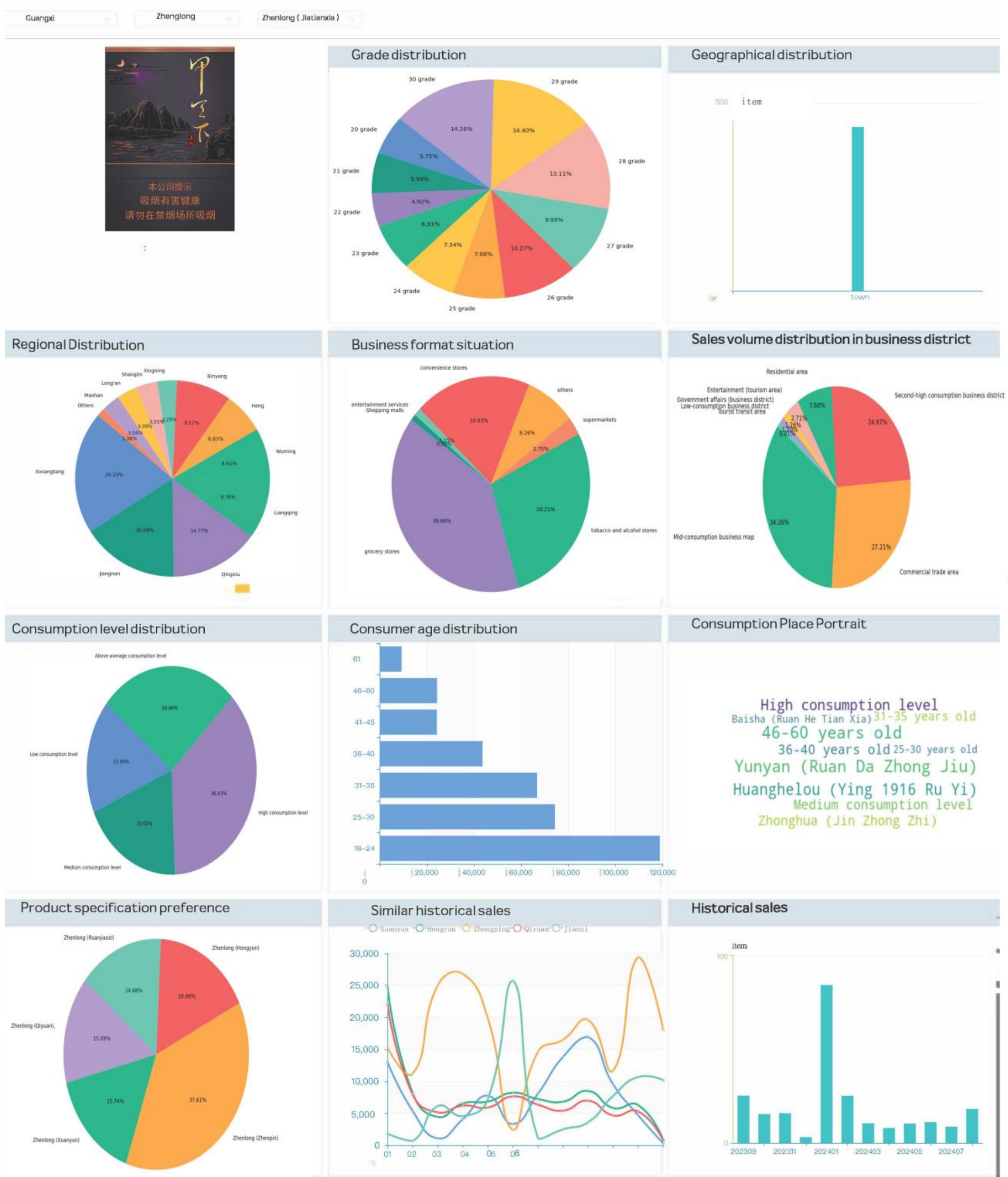

Figure 7: New Product Retail Terminal Selection Portrait.

from the selected terminal's grade distribution, geographical location distribution, regional distribution, business format conditions, business district sales distribution, consumption level distribution, consumer age distribution, consumption location portrait, product specification preferences and historical sales of similar products and other dimensions to help decision makers more intuitively understand the selected terminal distribution details, so as to further correct the selected target terminal and help new products grow rapidly.

4 SYSTEM APPLICATION

cigarette smart launch system developed in this paper has currently covered 100 cities in 31 provinces, autonomous regions, and municipalities across the country, and has been successfully applied to cigarette launch in many regions . Its effectiveness has also been verified by actual data. In 2021, during the application and promotion in the Yulin market in Guangxi, a precise launch strategy was implemented for Zhenlong (Haiyun Zhongzhi), and the launch precision rate increased by 14.8%. This achievement was demonstrated at the Guangxi Tobacco Commercial Agricultural Network Construction and Data Marketing On-site Conference and was further promoted and applied. In 2022, the system was applied to the listing cultivation of Zhenlong (Haiyun Zhongzhi) in the Jinan market in Shandong, which effectively helped the market penetration of new products and the improvement of consumer acceptance.Customer satisfaction increased by an average of 12%,achieved the cultivation goals, and promoted the brand's steady start and rapid growth in new markets. In 2023, in the application of Guigang market in Guangxi, the problems of unreasonable small-scale launch, low calculation accuracy, and seeking the optimal launch area were solved . The actual launch of multiple products such as Zhenlong (Haiyun), Zhenlong (Qiyuan), Zhenlong (Xuanyun), Zhenlong (Xiangyun), and Zhenlong (Zhenpin) were carried out, and the average launch accuracy was improved by 15.4%. The total time for launch was reduced from 1.5 days to half a day, and the launch efficiency was significantly improved. In the same year, the system was applied to the cultivation of the new product launch of Zhenlong (Jiatianxia) . The system was used to screen 1,500 target retail terminals across Guangxi, and finally 1,403 retail terminals successfully placed orders, with an accuracy rate of 93.5%. In the Yulin market, the sales-to-purchase ratio of Zhenlong (Jiatianxia) was effectively promoted, and the cumulative sales-to-purchase ratio reached 91.5%. The relevant application results were displayed and promoted at the Second Guangxi Regional Agricultural Network Construction and Data Marketing On-site Conference. In addition, the relevant results also attracted many units in the industry to come for exchanges and learning. Through the above practical verification, cigarette intelligent launch system has shown significant advantages in improving launch accuracy and work efficiency, providing strong data support and intelligent assistance for BB's business decision-making.

5 CONCLUSION

Based on consumer big data technology, this article successfully developed an cigarette smart launch system, and verified its significant effect in improving the efficiency and accuracy of cigarette market launch through practical applications. The system achieves a scientific assessment of cigarette sales potential by building a big data labeling system for end consumers and comprehensively using technical means such as big data analysis, correlation coefficients, and entropy methods . At the same time, various launch modes such as "by grade", "grade × label", "grade + label expansion", and new product selection have been constructed to effectively meet the diverse needs of different markets and products. In field applications in Jinan, Shandong, Guigang, Guangxi, Yulin, Guangxi and other markets, the system has significantly improved the accuracy and efficiency of cigarette launch, providing strong data support and intelligent assistance for tobacco business decision-making.In addition, the system is based on consumer behavior big data and sales potential evaluation model, and its core ideas and methods can be applied to other industries. For example, the retail, catering or fast-moving consumer goods industry only needs to redefine the label system according to its specific sales model and consumption scenario, and can accurately deliver through the "people, goods, and places" label system and big data analysis. In the future, with the continuous advancement of technology and the continuous evolution of the market environment, the system will continue to be optimized and upgraded to adapt to a wider range of application scenarios and more complex market demands, further promoting the intelligent development of the tobacco industry.

References

[1] FU Erju, LU Zhongqing, ZhANG Chao. Exploration of supply distribution model based on big data and artificial intelligence technology[J]. China Informatization, 2023, (07): 90-91.
[2] XU Hong, LIU Wenqian. Study on strategy for new cigarette products launch based on retailers' sales potential (G value) [J]. Acta Tabacaria Sinica, 2016, 22(01) :114-121.
[3] DENG Chao, LIU Song, WANG Ludi, GONG Qiang, GAO Lin, ZUO Shaoyan, GU Zuyi, LIANG Hailing. A method for building an intelligent cigarette delivery model based on a deep neural network[J]. Tobacco Science and Technology, 2021, 54(02):78 -83.
[4] MO Yuhua, TAN Qian, GU Yue, *et al.* Research on new cigarette launch strategy based on synthetic control method and machine learning [J/OL]. Journal of Chinese Tobacco Society, 1-9, 2024, 07-31. http://kns.cnki.net/kcms/detail/11.2985.TS.20240206.1400.002.html.
[5] Xue Xia Liang, Chao Deng, Bin Lu, *et al.* Design and application research on intelligent evaluation model of cigarette market operating status[C]//Proceedings of the International Conference on Computing, Machine Learning and Data Science. 2024, 1-7.
[6] XU Ruiqi, ZhOU Qiliang, BI Xunbo, *et al.* Research on chessboard-style precise placement of cigarettes in "stalls + business districts" [J]. Modern Business, 2022, (22): 3-6. DOI: 10.14097/j. cnki.5392/2022.22.022.
[7] WEI Taicheng, HUANG Feijie, ZhANG Xin, *et al.* Research and application of data mining based on artificial intelligence in cigarette delivery [C]//Implementing new development concepts and building a new development pattern-Collection of excellent papers of Guangxi Tobacco Society in 2021. Internet Research Center of Guangxi China Tobacco Industry Co., Ltd., 2022, 11. DOI: 10.26914/c.cnkihy.2022.080092.
[8] DENG Chao, LIANG Xuexia, CHEN Zhi, *et al.* Six portrait system based on cigarette consumption big data label system[J]. Acta Tabacaria Sinica, 2024, 30(1). doi:10.16472/j.chinatobacco.2022.063.
[9] Shoina M, Voukkali I, Anagnostopoulos A, *et al.* The 15-minute city concept: The case study within a neighbourhood of Thessaloniki[J]. Waste Management & Research, 2024, 42(8): 694-710.
[10] Mukaka M .Statistics corner: A guide to appropriate use of correlation coefficient in medical research.[J].Malawi Medical Journal, 2012, 24(3):69-71.
[11] GUO Xianguang. Improved entropy method and its application in economic benefit evaluation [J]. Systems Engineering - Theory and Practice, 1998, (12): 99-103.
[12] Gormley C, Tong Z. Elasticsearch: the definitive guide: a distributed real-time search and analytics engine[M]. " O'Reilly Media, Inc.", 2015.